KOÇ UNIVERSITY
College of Engineering

COMP 438/538: INTRODUCTION TO REINFORCEMENT LEARNING

Midterm 1

Fall 2026, 05/11/2026
Duration: 100 Minutes

Name: _____

ID: _____

Read These Instructions Carefully

- This exam contains 14 pages including this cover page and 5 questions. Check to see if any pages are missing. Put your initials on the top of every page, in case the pages become separated.
- By submitting this exam, you **agree** to fully comply with Koç University Student Code of Conduct, and accept any punishment in case of failure to comply. If you do not submit this exam on time, you will receive 0 credits.
- The exam is **closed book** and **closed notes**. You are **not** allowed to use any additional material including any electronic equipment such as computers and mobile phones. An equation sheet is provided on the last page.
- You are expected to be able to provide clear and concise answers. Gibberish will not receive any credit. Your answers need to be readable by a human, illegible writing is not gradable.
- Show the essential calculations or a short justification. Exact fractions or equivalent decimals are acceptable. Each question describes a separate problem; state names are local to their question. Terminal states have zero continuation value, and a reward shown on a transition is the total reward for that transition.
- Read each question carefully and make sure to follow instructions. The written instructions take precedence over an answer that the instructor might give you during the exam, unless the instructor makes a class-wide announcement.
- Do not write in the table below.

Question:	1	2	3	4	5	Total
Points:	12	22	32	20	14	100
Score:						

1. (12 points) Short answers.

   (a) (2 points) Consider two policies for a continuing service task. One produces reward 0.75 every step. The other repeats the rewards $0, 2, 0, 2, \ldots$ forever. Which has the higher long-run average reward per step, and why? No average-reward learning algorithm is required.

   (b) (4 points) A report compares MC and TD prediction on a fixed policy. It tries many step sizes and selects each method's best setting using the same episodes and seeds on which it reports final value error. It then uses the single lowest-error run to claim that the selected method is generally better. Identify two weaknesses and suggest a fix for each. No statistical test or numerical uncertainty calculation is required.

For each statement below, write **True** or **False** and give a one-line justification. Each is worth 2 points: 1 for the verdict and 1 for the justification. A verdict without a justification receives 0.

(c) (2 points) Given q^*, a greedy optimal policy can be obtained without the model, however, obtaining one from v^* by one-step lookahead needs the model.

(d) (2 points) If the current policy is not optimal, one greedy policy-improvement step using its exact action values increases the value of at least one nonterminal state.

(e) (2 points) The Monte Carlo prediction method we used in class can be applied directly to a continuing task that never terminates.

2. (22 points) An online tutoring app runs a practice session of exactly 20 questions. Before each question it chooses a difficulty: easy, medium or hard. After each question it sees only whether the answer was correct and how many seconds it took. The student's actual skill is not visible, and it changes as the student practises. After the 20th question the student takes a fixed 10-question test that the app does not choose.

   The designer's objective is to maximize the student's expected score on the fixed final test. No model of how difficulty affects skill, correctness or response time is supplied.

   (a) (4 points) What are the actions? What does the app observe after each question?

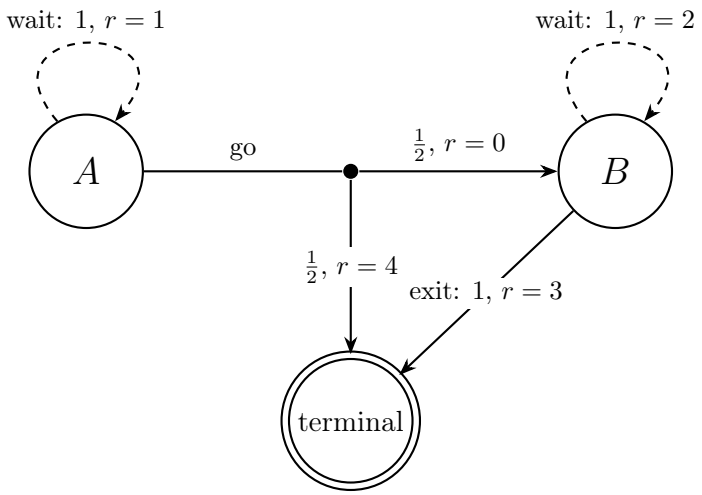
   (b) (6 points) Explain why the latest observation alone need not be a Markov state. Give two plausible histories ending in the same observation that could suggest different skill levels. Propose a history-based agent state for choosing the next difficulty, justify the information you retain, and mention one limitation. A justified approximation is sufficient; you do not need to prove that it is Markov.

(c) (6 points) Designer A rewards each correct practice answer with +1 and gives 0 otherwise. Designer B gives the final test score (0–10) after the session and 0 otherwise. Describe a plausible way A could encourage behaviour that does not improve the final test score. Which reward more directly matches the stated objective, and what makes learning from it difficult? You do not need to identify an optimal sequence of difficulties.

(d) (4 points) Is the task episodic or continuing? Should the number of questions remaining be part of the agent state? Why?

(e) (2 points) With no model of the student's responses or learning dynamics supplied, can you directly apply ordinary tabular value iteration to compute a policy? What information would be needed?

3. (32 points) There are two nonterminal states A and B, and a terminal state. The discount is $\gamma = \frac{1}{2}$. The figure shows every possible transition. The black dot shows the possible outcomes after choosing go. Each arrow gives the probability $\mathcal{M}(s, a, r, s')$ and the reward r of its outcome. All other outcomes have probability zero.



The policy π chooses wait and go with probability $\frac{1}{2}$ each at A, and always chooses exit at B. Every sweep (i.e. an application of a step of an iterative algorithm) below is **synchronous**: all new entries use the previous sweep's table. Terminal continuation is always zero.

(a) (6 points) Starting from the expected-return meaning of v^π, write the two Bellman equations for this particular policy. Show where both the policy probability and the transition probability enter at A. Explain why the remaining return after reaching B can be replaced by $v^\pi(B)$. Do not solve the equations.

(b) (4 points) Starting with $V_0(A) = V_0(B) = 0$, perform two synchronous policy-evaluation sweeps for π. Fill in the table, showing each backup.

k	$V_k(A)$	$V_k(B)$
0	0	0
1		
2		

(c) (8 points) Start again from $V_0(A) = V_0(B) = 0$. Perform two synchronous **value-iteration** sweeps, showing both action-backup values at each state and identifying the maximum. Do not continue the table from part (b).

Sweep	A: wait	A: go	$V(A)$	B: wait	B: exit	$V(B)$
1						
2						

(d) (8 points) Independently of parts (b) and (c), the exact values of the original policy are supplied: $v^\pi(A) = \frac{5}{2}$ and $v^\pi(B) = 3$. Compute q^π for each of the four state–action pairs. State the greedy improved policy and check that its selected q^π value is at least v^π at each state.

(e) (6 points) Let π' be the greedy policy from part (d). Part (d) checks that using π' for one step and then following π is at least as good as following π throughout: $v^{\pi}(s) \leq q^{\pi}(s, \pi'(s))$ at both states. Explain why using π' for two steps and then following π is at least as good as using it for one step. (Hint: apply the same inequality at the state reached after the first step.) What happens as the number of steps using π' grows, and what does this imply about $v^{\pi'}$ and v^{π}? Then state the main difference between a policy-iteration cycle and a value-iteration sweep.

4. (20 points) A fixed policy π produced these two complete episodes. Each arrow is labelled by the reward received on that transition. The model is unknown. Discount $\gamma = \frac{1}{2}$.

   - Episode 1: $A \xrightarrow{0} A \xrightarrow{2} B \xrightarrow{4}$ terminal
   - Episode 2: $A \xrightarrow{0} B \xrightarrow{2}$ terminal

   The actions are omitted because this question only evaluates π.

   (a) (7 points) Use **first-visit** Monte Carlo sample averages over both episodes to estimate $V(A)$ and $V(B)$. Then give the **every-visit** Monte Carlo estimate of $V(A)$. For each estimate, list the returns you included.

   (b) (7 points) Independently of part (a), initialize $V(A) = 2$ and $V(B) = 0$. With $\alpha = \frac{1}{2}$, apply TD(0) updates online, in time order, through **episode 1 only**. For each transition, give the target and the updated value.

Transition	Target U_t	Updated entry	$(V(A), V(B))$ after
$A \xrightarrow{0} A$			
$A \xrightarrow{2} B$			
$B \xrightarrow{4}$ terminal			

(c) (6 points) This part uses **new data** and $\gamma = 1$. The data are four complete episodes:

- twice: $C \xrightarrow{+1} D \xrightarrow{0}$ terminal;
- twice: $D \xrightarrow{+2}$ terminal.

Using the batch convention from class (repeated passes over this fixed data, increments summed over a pass with the table held fixed, and a small enough step size to converge), what do batch Monte Carlo and batch TD(0) converge to for $V(C)$ and $V(D)$? Which answer would you trust if you knew the process was Markov, and why might you doubt that here?

5. (14 points) This is a new independent episode, with $\gamma = \frac{1}{2}$ and time indices $0, 1, 2, 3$:

$$A \xrightarrow{2} B \xrightarrow{0} C \xrightarrow{4} \text{terminal}.$$

Hold the value table fixed throughout: $V(A) = 0$, $V(B) = 4$, $V(C) = 6$, and terminal value 0. Targets are for the visit to A at time 0; do not update the table between calculations.

(a) (4 points) Expand the return recursion twice and replace the unobserved return after the second transition with its value estimate. Write the resulting two-step target $G_{0:2}$ symbolically before inserting numbers.

(b) (6 points) Calculate the one-step target $G_{0:1}$, the two-step target $G_{0:2}$ and the full return G_0. For the full return, indicate why there is no additional bootstrap term.

(c) (1 point) If this episode arrives one transition at a time, at which time index does the two-step target for A first become available? Must the learner wait until termination here?

(d) (3 points) A student sets $\lambda = 1$ in the finite episodic forward λ-return to remove bootstrapping, then concludes that the target must be more accurate than TD(0). Assess both claims briefly.

Several Equations (semantics and context are not given on purpose)

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}, \qquad G_t = R_{t+1} + \gamma G_{t+1}, \qquad G_T = 0$$

$$v^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \qquad q^\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

$$v^\pi(s) = \sum_a \pi(a \mid s)\, q^\pi(s,a)$$

$$q^\pi(s,a) = \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, v^\pi(s')\big]$$

$$v^*(s) = \max_a \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, v^*(s')\big], \qquad \pi^*(s) \in \arg\max_a q^*(s,a)$$

$$V_{k+1}(s) = \sum_a \pi(a \mid s) \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, V_k(s')\big]$$

$$V_{k+1}(s) = \max_a \sum_{s',r} \mathcal{M}(s,a,r,s')\big[r + \gamma\, V_k(s')\big]$$

$$V(S_t) \leftarrow V(S_t) + \alpha\big[U_t - V(S_t)\big]$$

$$U_t = G_t, \qquad U_t = R_{t+1} + \gamma V(S_{t+1})$$

$$V(s) \leftarrow V(s) + \frac{1}{N(s)}\big[G_t - V(s)\big], \qquad S_t = s$$

$$G_t^\lambda = (1-\lambda) \sum_{n=1}^{T-t-1} \lambda^{n-1} G_{t:t+n} + \lambda^{T-t-1} G_t, \qquad 0 \le \lambda \le 1, \quad t < T$$